

MERT KARACA

Industrial Engineering Student | Treasury, Financial Modeling & Operations Planning
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SUMMARY

Final-year Industrial Engineering student at Yaşar University with internship experience in banking (Treasury & Balance Sheet Management) and industrial manufacturing (Planning). Experienced in liquidity modeling, Monte Carlo simulation, inventory and investment analysis, and ERP workflows. Seeking opportunities in treasury balance sheet management, financial modeling, and operational planning.

EXPERIENCE

Treasury / Balance Sheet Management Intern | Akbank T.A.Ş. | Jul 2026 - Aug 2026

- Shadowed the liquidity team daily to follow how TL and foreign currency cash positions were tracked, closed, and funded.
- Calculated daily swap rates for liquidity management; tracked bond yields and market data using Bloomberg Terminal.
- Followed daily ALM pricing decisions, including how branch loan and deposit requests were evaluated against internal funding rates.
- Helped review daily market reports and data tables prepared for morning management meetings.
- Gained exposure to how Balance Sheet Management works with the Treasury Trading and Treasury Marketing desks.

Planning Department Intern | Doğu İklimlendirme | Jul 2025 - Aug 2025

- Examined the ERP system to understand how production planning and daily data flows are handled on the shop floor.
- Collected, analyzed, and reported daily operational data for the planning team.
- Supported data collection and process improvement steps as part of OPEX (Operational Excellence) work.
- Updated operation times and planning parameters to support routine decision-making.

EDUCATION

B.Sc. Industrial Engineering | Yaşar University, İzmir | Sep 2022 - Present | GPA: 2.40 / 4.00

ACADEMIC PROJECTS

Liquidity Buffer Sizing Model | IE 4830 Internship Project, Akbank T.A.Ş. / Yaşar University | Individual | 2026

- Built a quantitative model to determine the optimal next-day overnight liquidity buffer under cash flow uncertainty.
- Analyzed 53 months of banking sector deposit data from the CBRT EVDS system to estimate cash flow volatility.
- Ran a 100,000-scenario Monte Carlo simulation in Python and Excel to minimize emergency funding and idle-cash costs; compared results with the analytical Newsvendor model.
- Tested how the optimal buffer changes under higher funding interest rates and increased market volatility.

Airport Departure Process Simulation & Optimization | System Simulation (Arena), Yaşar University | Group | 2026

- Built an Arena simulation of a regional airport's departure process (check-in, bag drop, security, walk to gate), including ID-check failures and extra searches.
- Fitted distributions to time-study data and validated the fit with Chi-Square tests.
- Analyzed queue lengths, waiting times, and staff utilization to identify the main bottleneck; compared standard and expedited security lines and recommended the best 2026 flight schedule scenario to reduce waiting times.

Inventory Optimization (Q, r) Policy | Production and Service Systems Planning, Yaşar University | Group | 2026

- Modeled a continuous review inventory system under uncertain demand and lead time in Python; calculated safety stock, service levels, and total inventory costs; compared backorder and lost-sales cases.

Engineering Economics - Investment Appraisal | Engineering Economics, Yaşar University | Group | 2025

- Evaluated investment options using Net Present Worth (NPW) and incremental Internal Rate of Return (IRR); built cost-based cash flow models to support the final equipment selection decision.

TECHNICAL SKILLS

Programming & Data: Python (Beginner-Intermediate), SQL (Beginner), Machine Learning (Beginner), MS Excel (Intermediate)

Tools & Methods: Bloomberg Terminal, Arena Simulation, ERP Systems, Monte Carlo Simulation, Newsvendor Model, Inventory Management, Chi-Square Testing, NPW / IRR Analysis

CERTIFICATIONS

- ISO 9001 Quality Management Systems Training (2026) - Turkish Quality Association (KalDer)
- Microsoft Excel: Beginner to Advanced (2025) - Udemy
- Digital Financial Literacy Training (2024) - Habitat Association

LANGUAGES

English: Upper-Intermediate (B2) | Spanish: Elementary (A1)